EEE223 CIRCUIT ANALYSIS LABORATORY-1

EXPERIMENT 1

MEASUREMENT OF VOLTAGE, CURRENT AND RESISTANCE

1. PURPOSE OF EXPERIMENT

Measurement of current, voltage, and resistive quantities by use of an AVO meter and the experimental verification of Kirchhoff's current and voltage Laws.

2. THEORY

AVO meter (Multimeter) is an instrument for measuring the values of current, voltage, resistance. In case of appropriate ranges and modes of operation are selected, it is possible to get a correct measurement. Otherwise, either the meter could be damaged, or the measurement will be incorrect. The multimeter is shown in Figure1.



Figure 1. *Multimeter*

2.1. *AVO meter in current mode:*

In order to use the AVO meter as an ammeter, one has to observe the following directions:

*When measuring current with the AVO meter, the positive (red) probe of the AVO meter must be connected to the port reserved for current measurement.

*The ampermeter should be connected in **series** to any network since its internal resistance is supposed to be zero.

*In direct current (dc) circuits, the current flow direction is important. The ammeter should be connected in such a way that the current enters the instrument from the (+) terminal. Otherwise, the pointer will deflect in the reverse direction.

As an example, refer to Figure 2. If one wants to measure the current I2 in the Figure, first all the switches must be properly set, so that the meter acts as an ammeter. Then the meter terminals are connected to the branch is in series. Other currents in other branches could be measured in a similar manner.

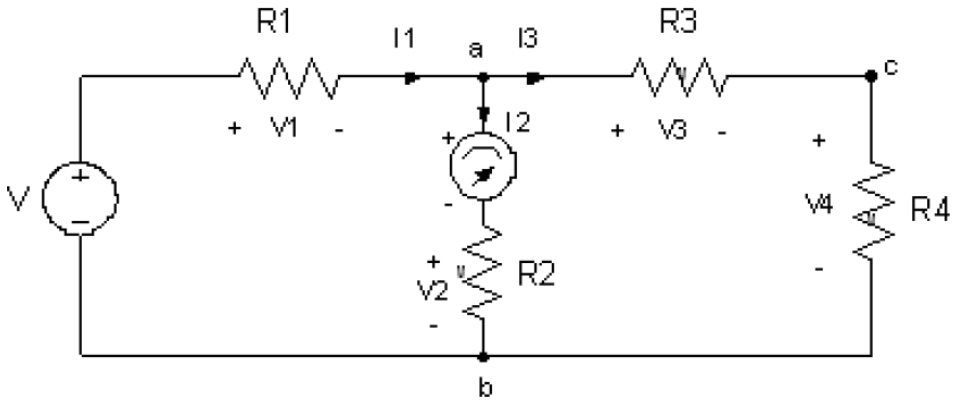


Figure 2. *Measurement of current*

2.2. AVO meter in voltage mode

In order to use the AVO meter as an voltmeter, one has to observe the following directions;

*As in the ammeter case, one has to set the switches on the meter properly, so that the meter acts as a voltmeter.

*The voltmeters are always connected to the network in **parallel**, as its internal resistance is supposed to be infinite.

*In alternating current (ac) circuits, the connection of the terminal polarities is not important. But for dc measurements one has to pay attention for the appropriate connections. As an example, if voltage V3 is to be measured in Figure 4, the meter terminals must be connected as shown below.

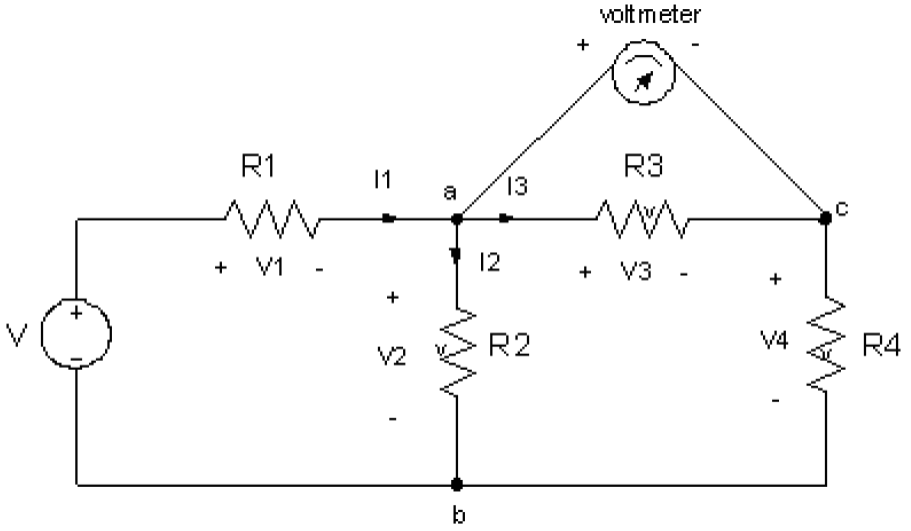


Figure 3. *Measurement of voltage*

2.3. AVO meter in resistance mode

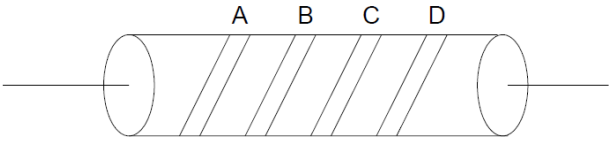
The switches on the meter should be selected appropriately. The AVO meter's probes should be shorted out so that zero adjust can be made. Ohmmeter is connected across the terminals of a resistor, both terminals of which should be free.

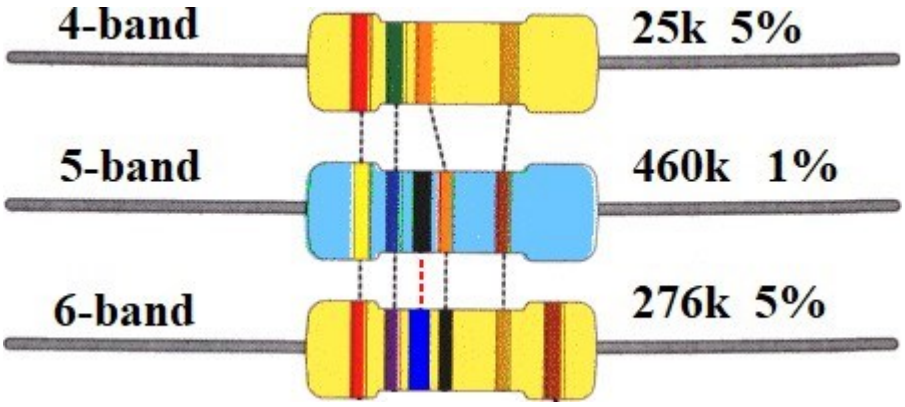
2.4. Resistance Color Coding

The most widely used resistances are carbon type. There exists three- or four-color bands on the resistance body, indicating the ohmic values and tolerances. In order to achieve a correct reading, one has to hold the resistance in such a way that the bands are close to the left-hand side, then the ohmic value is given by the formula below.

Color	Value
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Gray	8
White	9
Gold	-1 (only for band D)
Silver	-2 (only for band D)

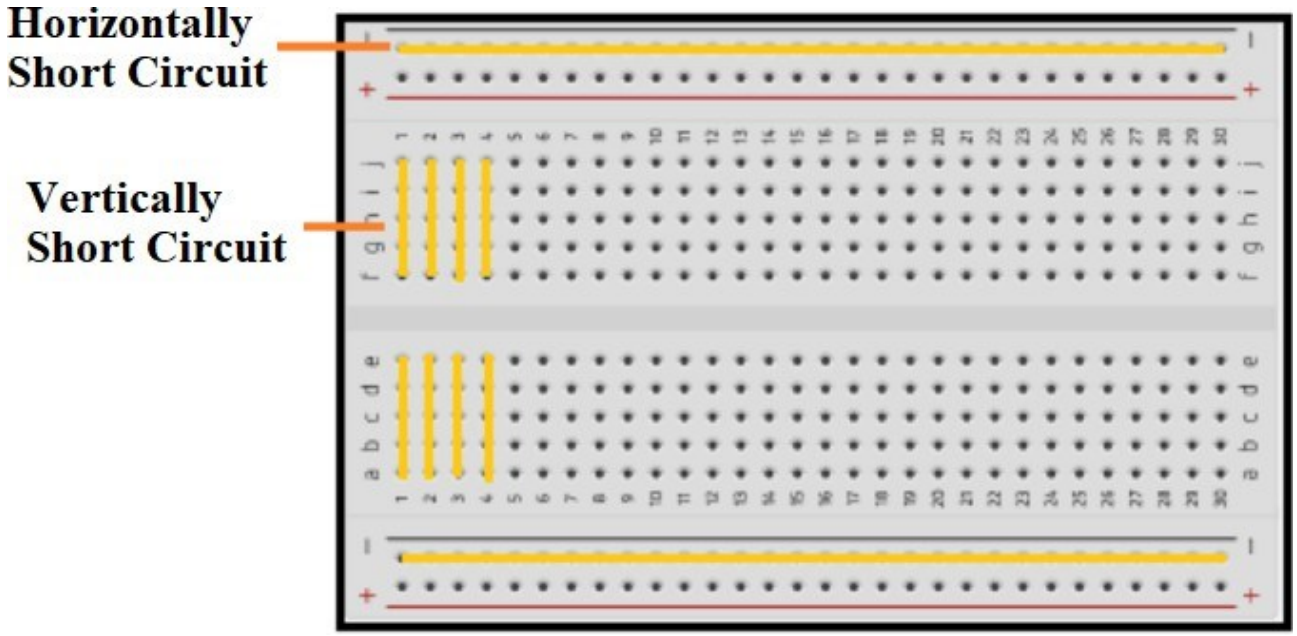
$R=ABx10^{C} \pm D$ ohms





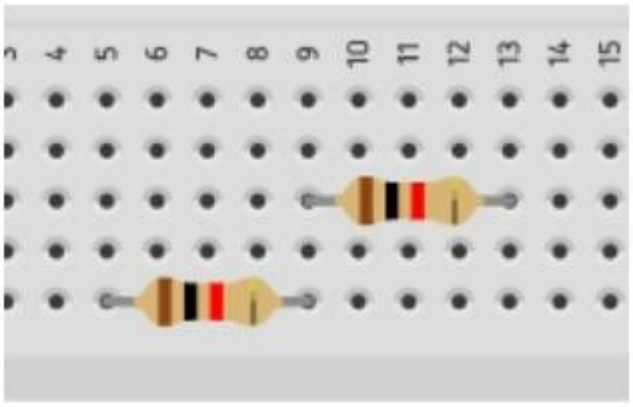
2.5. Breadboard Connections

White plastic is the material that is used to create breadboards. Today, most of the breadboards are Solderless breadboards. We can directly plug in the electronic components and connect it with the external power supply. The top and bottom holes of a row in a breadboard are connected horizontally, and the center part is connected vertically. Electrical connections of the breadboard holes is shown below:

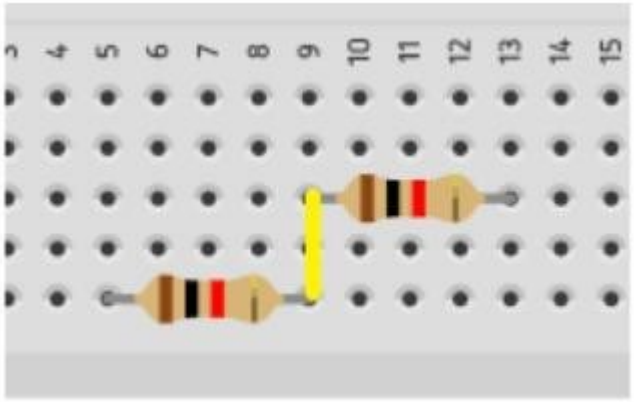


Example: Rows **a**, **b**, **c**, **d** and **e** in column **1** are short-circuited among themselves.

It means a single horizontal line of a breadboard has the same connection. It is because the metal strip underneath the breadboard at the top and bottom are connected horizontally. Hence, it provides the same connection in a row. The two top and bottom parts of a breadboard are generally used for power connections. The vertical connection of the center part means a single vertical line in a breadboard provides the same connection. It is useful when we need to connect the different components in series. For example, let's connect two resistors in series. These two resistors can be connected in series in different ways, as shown below:

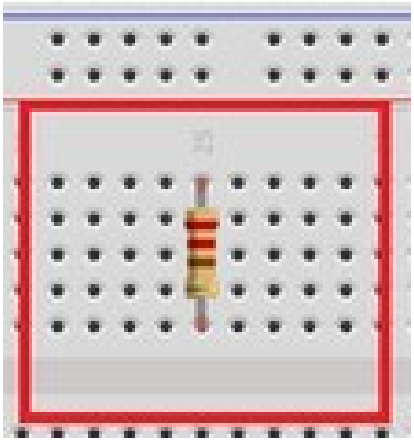


It is because the metal strips underneath the breadboard at the center are connected vertically. Hence, it provides similar connectivity through a particular column, as shown below:

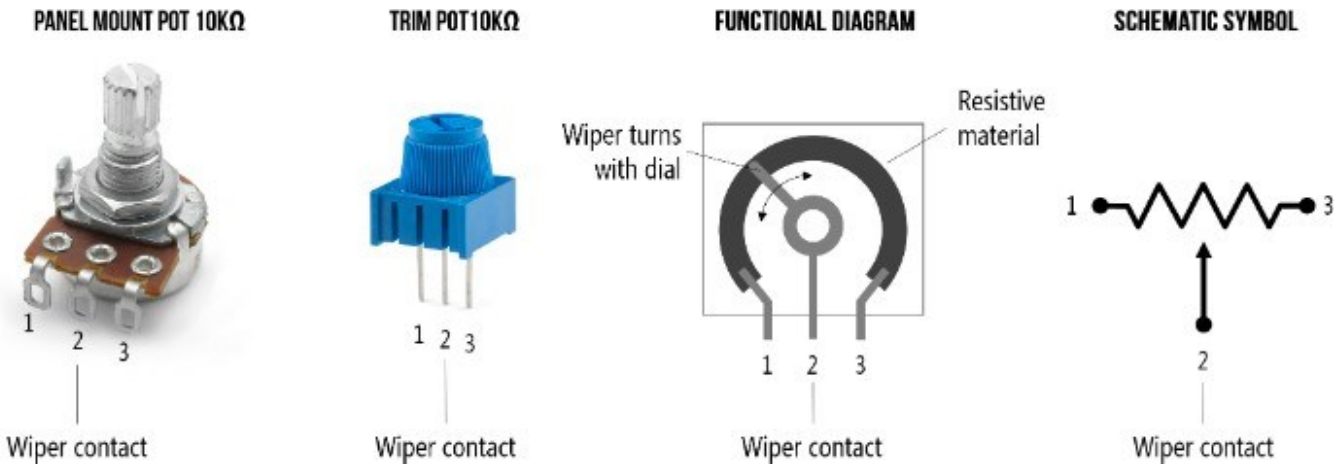


The connection between two different components can be created by inserting a lead in common. For example, a jump wire that acts as a connection between the LED and battery terminal can be connected in any hole in the same vertical line.

The Example of ***WRONG CONNECTION*** Is Given Below**!!**

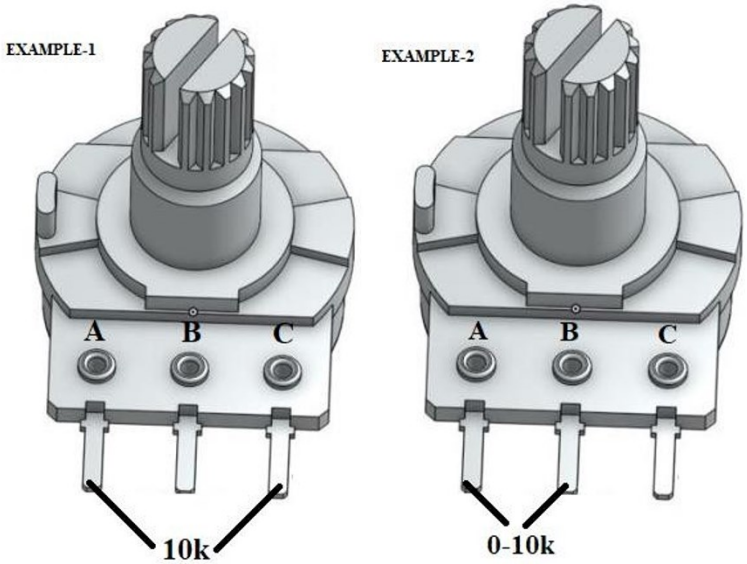


2.6. *Potentiometer Connections*



If the 10kΩ potentiometer is connected to the circuit using terminals A and C, the potentiometer is used as a 10kΩ resistor (**EXAMPLE-1**)

If the 10kΩ potentiometer is connected to the circuit using terminals A and B, the potentiometer is used as a variable resistor. Variable between 0-10kΩ (**EXAMPLE-2**)



3. PRELIMINARY WORK

3.1. Calculate the values of currents I1, I2, I3 and the voltages V1, V2, V3, V4 in Figure 4. Show the meter connections for these measurements. Fill in the table below.

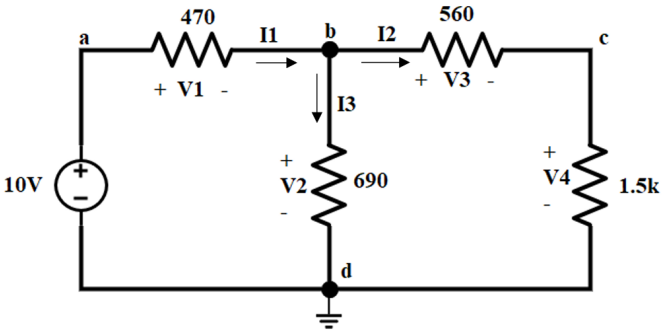


Figure 4

I1 (mA)	I2 (mA)	I3 (mA)	Rad (Ω)	V1 (V)	V2 (V)	V3 (V)	V4 (V)

3.2. For the circuit given in Figure 5 calculate the voltage and current values on R2 according to the R1 values given in the table below and fill in the relevant table.

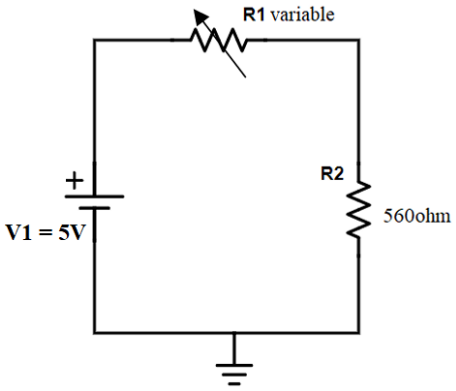


Figure 5

	Variable resistor (4kΩ)	Variable resistor (8kΩ)
Current (mA)		
R2 Voltage (V)		

3.3. Find the equivalent resistance Rab between points a and b for the circuit in Figure 6.

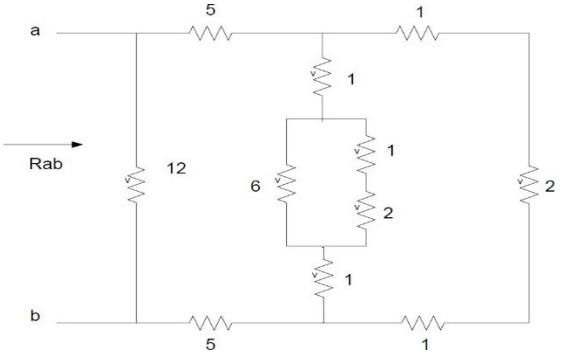


Figure 6

3.4. For the circuit shown in Figure 7. the value of R2 is required to be measured. Why does not such a connection give the correct value?

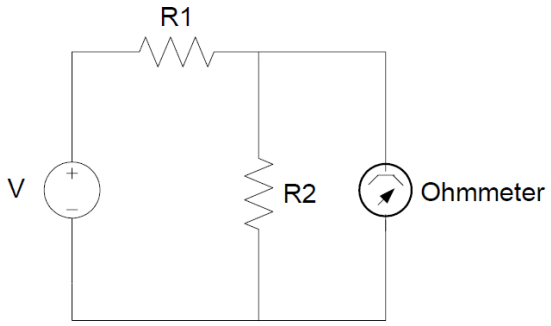


Figure 7

3.5. Figure 8. shows a carbon resistor.

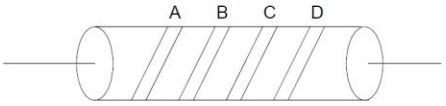


Figure 8

For each of the resistance values given, write down the correct color bands.

Resistor Values	**Color Codes**			
	A	**B**	**C**	**D**
100Ω ±10%				
120Ω ±5%				
220Ω ±20%				
330Ω ±10%				

4. SIMULATION WORK

4.1. Set up the circuit given in Figure 9 in simulation and measure the values of currents I1, I2, I3 input resistance (Rad) and the voltages V1, V2, V3, V4.

I1 (mA)	**I2 (mA)**	**I3 (mA)**	**Rad (Ω)**	**V1 (V)**	**V2 (V)**	**V3 (V)**	**V4 (V)**

4.2. Set up the circuit given in Figure 10 in simulation and fill in the table below according to the desired values for the adjustable resistor.

	Variable resistor (4kΩ)	**Variable resistor (8kΩ)**
Current (mA)		
R2 Voltage (V)		

5. EXPERIMENTAL WORK

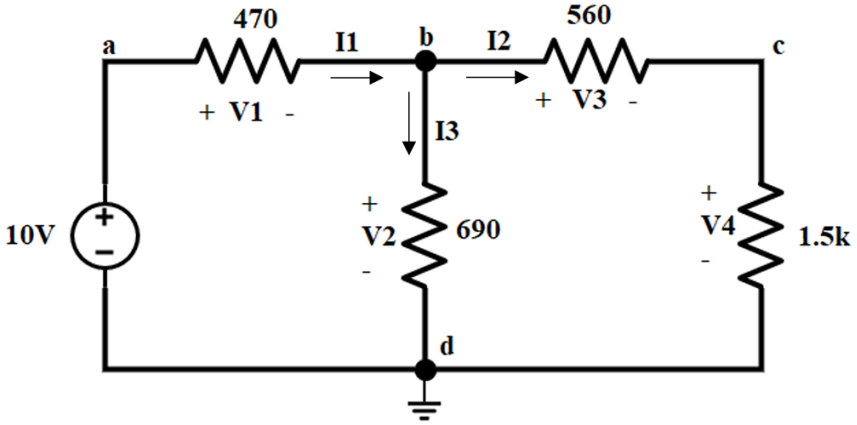


Figure 9. *Circuit-1 for the experimental work*

Set up the circuit given in figure 9.

5.1. Measure the values of each resistor with an ohmmeter before connecting them to the circuit. Write the relevant values in the table below.

5.2. Measure the values of currents I1, I2, I3 and the voltages V1, V2, V3, V4 by use of an AVO meter. Write the relevant values in the table below.

5.3. Disconnect the supply. Measure the value of input resistance of the circuit seen from supply side (R_{ad}). Write the value in the table below.

I1 (mA)	I2 (mA)	I3 (mA)	V1 (V)	V2 (V)	V3 (V)	V3 (V)	V4 (V)	Rad (Ω)	R1(Ω)	R2(Ω)	R3(Ω)	R4(Ω)

Set up the circuit given in figure 10:

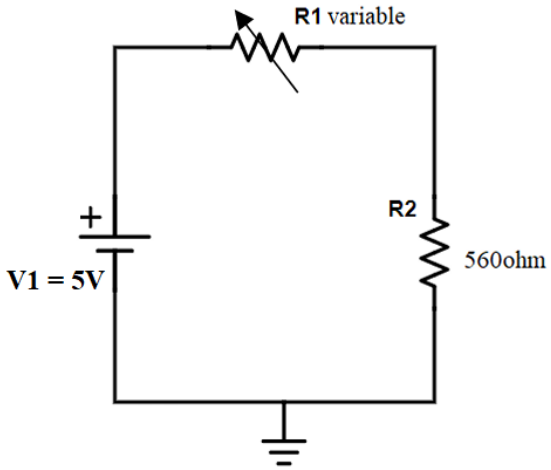


Figure 10. *Circuit-2 for the experimental work*

5.4. For the circuit given in Figure 10: Adjust the variable resistor to 4kΩ and measure the current and voltage on the R2 resistor. Then, Adjust the variable resistor to 8kΩ and measure the current and voltage on the R2 resistor. (V1=5V)

	Variable resistor (4kΩ)	Variable resistor (8kΩ)
Current (mA)		
R2 Voltage (V)		

5.5. For the circuit given in Figure 10 calculate the resistor R1 for which the circuit current will be 2.5mA theoretically. Then adjust the potentiometer to the resistance value you determined and connect it to the circuit. After the circuit is complete, measure the current and the voltage across the resistor R2. Fill in the table below.

R1 value (Ω) =	**Calculated Current (mA)**	**Calculated R2 Voltage (V)**
	Measured Current (A)	**Measured R2 Voltage (V)**

Equipment and Components

Cihaz	**Değer ve Adet (#)**
DC Power Supply	1
AVO meter	1
Resistors	470Ω(#1), 560Ω(#1),690Ω(#1),1.5kΩ(#1)
Variable resistor	10kΩ (#1)